



Post-marketing safety evaluation of the live-attenuated chikungunya vaccine (IXCHIQ)

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ABSTRACT

Background: IXCHIQ (VLA1553) is a licensed live-attenuated chikungunya vaccine. A post-marketing safety review, including the EU Article 20 referral to the European Medicines Agency (EMA) and its Pharmacovigilance Risk Assessment Committee (PRAC), was initiated following early real-world use during the 2024–2025 La Réunion outbreak. This analysis integrates global serious adverse event (SAE) data with major regulatory assessments to inform clinicians and policymakers.

Methods: Serious adverse events reported from 9 November 2023 to 31 August 2025 were evaluated across major pharmacovigilance systems (EudraVigilance, Vaccine Adverse Event Reporting System, Canada Vigilance, MHRA) and cases received by the marketing authorisation holder (MAH).

Results: By 31 August 2025, approximately 55,900 doses had been administered worldwide, yielding 35 SAEs (6.3 per 10,000 doses). One-third of doses (33%) were administered in adults ≥ 65 years, who accounted for 77% of SAEs, reflecting the older, multimorbid population initially targeted during outbreak response in La Réunion. The median age of persons with SAEs was 73 years; 91% had comorbidities. Most events occurred within 7 days and included systemic chikungunya-like reactions (15), neurological (13), cardiac (3), and renal (3) manifestations. Three fatalities occurred in older men with multimorbidity; one case of vaccine-strain encephalitis was considered probably related to vaccination. No manufacturing or batch-related issues were identified.

Abbreviations: ACIP, Advisory Committee on Immunisation Practices; ANVISA, Agência Nacional de Vigilância Sanitária (Brazilian Health Regulatory Agency); CDC, Centres for Disease Control and Prevention; CHIKV, Chikungunya Virus; CHMP, Committee for Medicinal Products for Human Use; CISA, Clinical Immunisation Safety Assessment; CNAM, Caisse Nationale d'Assurance Maladie (French National Health Insurance Fund); CSF, Cerebrospinal Fluid; EC, European Commission; EEA, European Economic Area; EMA, European Medicines Agency; FDA, Food and Drug Administration; HAS, Haute Autorité de Santé (French National Health Authority); ICSR, Individual Case Safety Report; LAV, Live-Attenuated Vaccine; MAH, Marketing Authorisation Holder; MHRA, Medicines and Healthcare products Regulatory Agency; NITAG, National Immunisation Technical Advisory Group; PLOS, Public Library of Science; PRAC, Pharmacovigilance Risk Assessment Committee; PSURs, periodic safety Update Reports; RT-PCR, Reverse-Transcription Polymerase Chain Reaction; SAE, Serious Adverse Event; SmPC, Summary of Product Characteristics; UK, United Kingdom; US, United States; USPI, United States Prescribing Information; VAERS, Vaccine Adverse Event Reporting System; WHO, World Health Organization.

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Conclusions: Early post-marketing SAEs occurred mainly in older adults with multimorbidity, reflecting reactivity patterns expected with live-attenuated vaccines, while serious events were uncommon in adults <65 years relative to doses administered. Regulatory reviews concluded that IXCHIQ's overall benefit–risk balance remains favourable within authorised and nationally recommended populations, and product information was updated to reinforce warnings, precautions, and contraindications.

1. Introduction

Chikungunya virus (CHIKV) is an *Aedes* mosquito-borne alphavirus that has caused repeated outbreaks across Africa, Asia, and the Americas, with imported and autochthonous cases increasingly reported in Europe [1,2]. As of 31 December 2025, Metropolitan France had reported 788 acquired chikungunya cases and Italy 384 [2]. Although often self-limiting, chikungunya disease can lead to substantial morbidity, including prolonged arthralgia, neurological complications, and severe or even fatal outcomes in vulnerable groups such as neonates, older adults, and individuals with comorbidities [3–5]. The global expansion of competent vectors emphasises the need for effective prevention strategies, particularly for travellers and populations in endemic regions.

IXCHIQ®, a live-attenuated monovalent chikungunya vaccine derived from the La Réunion LR2006-OPY1 strain, was authorised by the United States (US) Food and Drug Administration (FDA) in 2023 for adults aged 18 and older who are at increased risk of exposure to CHIKV. It was subsequently approved by the European Medicines Agency (EMA) and Health Canada in 2024, by the United Kingdom (UK) Medicines and Healthcare products Regulatory Agency (MHRA) in 2025, and by Brazil's Agência Nacional de Vigilância Sanitária (ANVISA) in 2025 (Table 1). In addition, the EMA and Health Canada have extended approval to include adolescents aged 12–17, with reviews of this label extension by other regulatory agencies ongoing. In 2026, Valneva announced that it would voluntarily withdraw the US biologics license application (BLA) and Investigational New Drug (IND) application for IXCHIQ® following the FDA's suspension of the US license in 2025. Licensure was based on immunogenicity endpoints, with the Phase 3 clinical trial data demonstrating high immunogenicity and an acceptable safety profile, including more than 3600 participants [6,7].

Following rollout, IXCHIQ® entered post-marketing surveillance systems in multiple regions. By 31 August 2025, approximately 55,900 doses had been administered worldwide, including substantial use during the escalating 2024–2025 La Réunion epidemic. In March/April 2025, the French National Immunisation Technical Advisory Group (NITAG), Haute Autorité de Santé (HAS), recommended emergency vaccine deployment to those at the highest risk of severe wild-type disease, namely adults ≥65 years and adults aged 18–64 with comorbidities [8]. This strategy prioritised patients most vulnerable to severe chikungunya; however, these groups were also more susceptible to potential adverse outcomes from a live-attenuated vaccine (LAV) as also recognised for the live-attenuated yellow fever vaccine, where age ≥ 60 years is a precaution. In addition, emergency campaign conditions led to instances of off-label use, including administration in immunocompromised individuals and older adults with unstable multimorbidity.

After early serious safety reports, local and international authorities implemented precautionary measures. In Europe, HAS temporarily

halted use in individuals aged ≥65 years [8], and in May 2025 the EC initiated an Article 20 referral, mandating PRAC to review cumulative evidence and reassess the vaccine's benefit–risk balance. Temporary age-related measures were applied during this review. Following PRAC's assessment and CHMP's opinion, the EC maintained IXCHIQ's indication without an upper-age restriction but strengthened the product information with additional warnings, precautions, and adverse reaction information [9,10]. In parallel, the FDA and CDC reviewed VAERS data and MAH submissions; the FDA updated US label warnings and precautions in early August 2025 and subsequently temporarily suspended the US license pending further evaluation [11,12]. Health Canada and the MHRA conducted similar reviews and, following global reports of serious adverse events, temporarily restricted IXCHIQ use in people aged ≥65 years; as of November 2025, the vaccine remains available for individuals aged 18 and above in the UK, while the Health Canada review is still ongoing (23 October 2025) [13].

The objective of this manuscript is to provide a post-marketing safety review of IXCHIQ with data up to 31 August 2025, with particular attention to the EU Article 20 referral procedure and its outcomes, as this regulatory mechanism has delivered the most comprehensive and methodologically rigorous appraisal to date. We transparently consolidate all available safety findings, including pre-licensure clinical trial benchmarks and product-quality data, to offer a clear evidentiary basis for clinicians and policymakers interpreting early post-authorisation safety reports—showing what was reported, how it was evaluated, and what conclusions were reached across systems and regulators.

2. Methods

2.1. Data sources

Post-marketing safety data were drawn primarily from Individual Case Safety Reports (ICSRs) available in EMA's EudraVigilance database, including case analyses coordinated in France (and La Réunion) by the Caisse Nationale d'Assurance Maladie (CNAM) and HAS, as well as cases received by the MAH directly and submitted to EudraVigilance.

US post-marketing safety information was obtained from VAERS, the national spontaneous reporting system jointly managed by CDC and FDA. Reports were identified using IXCHIQ product terms, classified for seriousness using EU regulatory criteria [14], and supplemented with additional clinical details and laboratory data (including CDC RT-PCR or serology when available). CDC's ACIP used these data to inform interim clinical guidance, and FDA conducted benefit–risk reassessments and managed US labelling and risk communication, following established CDC vaccine-safety workflows [11]. Because clinical trial data provide the pre-licensure safety benchmark, we briefly summarise IXCHIQ® trial safety results relevant to this review.

2.2. Analytical approach

As standard practice, all available information was entered into the MAH's global safety database exactly as received directly or retrieved from the authorities' websites. Reports were received through EudraVigilance, VAERS, Health Canada, MHRA, or directly by the MAH, providing a consolidated global dataset for descriptive analysis. Subsequently, the MAH performed a medical assessment, including a causality assessment. All serious cases received by the MAH were assessed individually. Clinical narratives, comorbidities, concomitant medications,

Table 1
Regulatory status of IXCHIQ as of October 2025.

Country/region	Date of initial approval	Current indication
USA	09 Nov 2023	License suspended 22 Aug 2025
EU	28 Jun 2024	Indicated for ages 12 and older
Canada	20 Jun 2024	Indicated for ages 12 and older
UK	04 Feb 2025	Indicated for ages 18 and older
Brazil	14 Apr 2025	Indicated for ages 18 and older

and time to onset were reviewed by the MAH.

To ensure consistency across all subsequent analyses, we pre-specified a data-lock date of 31 August 2025 and consolidated all serious post-authorisation safety reports from multiple reporting systems into a single, integrated analytic dataset. We provided a quantitative overview of the post-marketing case set before detailing the EU PRAC/CHMP assessment and outcome and summarising key regulatory interactions relevant to ongoing pharmacovigilance and clinical guidance.

3. Results

3.1. Summary of the clinical trial safety results IXCHIQ

To contextualise the post-marketing findings, we first summarise safety data reported during clinical trials, which serve as the foundation for comparison with real-world experience. Across the pre-licensure programme (3520 IXCHIQ recipients in Phase 1 and 3 studies, including 346 older adults aged 65 and above), a single dose of IXCHIQ was considered safe and well tolerated. In pooled Phase 1 and 3 analyses of the licensed dose regimen, solicited systemic AEs occurred in 50.9% of vaccinees versus 26.9% with placebo; arthralgia in 16.7% versus 4.8%; medically attended AEs in 12.4% versus 11.3%; AEs of special interest in 0.3% versus 0.1%; and SAEs in 1.5% versus 0.8%. Two SAEs (~0.1%; 5.7 per 10,000) were assessed as vaccine-related—myalgia (classified as serious based on SAE criteria rather than symptom intensity) and SIADH—and both resolved without sequelae [15]. The majority of solicited systemic events began within the first few days and resolved within ≤5 days [15]. The most common vaccination site reactions were tenderness (11.0%) and pain (6.1%), and the most common systemic adverse reactions included headache (31.9%), fatigue (29.5%), myalgia (23.8%), arthralgia (16.7%), fever (13.8%), and nausea (11.4%) [15,16]. Together, these data indicate that while transient systemic reactogenicity is common, serious vaccine-related events were rare, and older adults mounted antibody responses comparable to those of younger adults [15,17–20].

3.2. Overview of the post-marketing safety dataset (data cut-off 31 august 2025)

As of 31 August 2025, the MAH's global pharmacovigilance system had recorded 35 serious ICSRs related to IXCHIQ vaccination among approximately 55,900 doses administered worldwide (23,200 in the US; 31,500 in the EU/EEA (including ~31,000 in France, including La Réunion); and 1200 in the UK). This corresponds to an overall reporting rate of 6.3 SAEs per 10,000 doses (35/55,900).¹ Among adults aged ≥65 years, 27 SAEs were reported after 18,500 doses (14.6 per 10,000), compared with 8 SAEs after 37,400 doses in those <65 years (2.1 per 10,000). The median age of cases was 73 years (range 28–89), with 27/35 (77%) aged ≥65 years and 27/35 (77%) male. Sex-specific denominator data for vaccinated individuals were not available; therefore, sex-stratified SAE reporting rates could not be calculated. Comorbidities were present in 32/35 cases (91%), including 29 individuals (83%) with multiple chronic conditions. Common comorbidities included hypertension, diabetes, cardiovascular disease, chronic respiratory disease, renal impairment, and immunological disorders. Most serious ICSRs were reported from France, including an emergency deployment of

vaccination in La Réunion. Reports originated from France ($n = 21$, including La Réunion), the US ($n = 10$), Canada ($n = 2$), Austria ($n = 1$), and the UK ($n = 1$). Outcome information at the last follow-up was available for all 35 individuals. Eleven had fully recovered, two were still recovering, eight had partially recovered (some symptoms resolved while others persisted), five had not recovered, and three had a fatal outcome (described later); in six additional cases, the reporting clinician documented the outcome as unknown or not yet determined. As with other passive surveillance systems, “not recovered” or “unknown” often reflects limited or no follow-up rather than confirmed ongoing symptoms; thus, true long-term recovery status cannot be determined for all cases. The data of all reported SAEs is available in the Supplementary Table 1.

Among the 35 serious cases, 15 (43%) systemic “chikungunya-like adverse reactions” (CLARs) were identified. These are defined as fever in combination with additional symptoms, such as arthralgia/arthritis, malaise, myalgia, neurological or cardiac events, occurring within 30 days post-vaccination. For the detailed definition and included symptoms, see supplementary materials. Neurological events occurred in 13/35 (encephalitis/encephalopathy, confusional states, vertigo, diplopia, behavioural disturbance, meningismus). Cardiac events were reported in 3/35 (myocardial infarction, tachyarrhythmia, pericardial effusion), and renal events in 3/35 (acute kidney injury, including acute renal failure, sometimes in the context of systemic decompensation). Other presentations included haematological (thrombocytopenia, haemolysis), dermatological, and endocrine/metabolic complications. Most events occurred within the first seven days after vaccination, consistent with the expected early reactogenicity window of an LAV [15]. It is important to note that categories listed are not mutually exclusive; totals exceed the sum when cases exhibited multi-system involvement.

3.3. PRAC assessment may 2025

Following early reports of SAEs in older vaccinees, the PRAC convened an oral explanation on 6 May 2025 and introduced temporary age-related restrictions on 7 May. The EC subsequently triggered an Article 20 referral on 8 May. The MAH submitted responses by 26 May, and the PRAC—after consultation with the Emergency Task Force—adopted its recommendation on 10 July 2025. As part of the referral, the PRAC reviewed 74 ICSRs from EudraVigilance (cut-off 25 May), including 28 serious cases after de-duplication: 18 from France/La Réunion, eight from the US, one from Canada, and one from Austria. Most cases occurred in men ≥65 years (22/28; 79%), almost all with comorbidities (26/28; 93%). Causality was assessed as probable/consistent in seven cases (including the fatal encephalitis with vaccine-strain detection), possible/indeterminate in 15, unlikely in one, and unclassifiable in five. Three cases were fatal, and 15 required hospitalisation. Based on this assessment, the PRAC concluded that the increased risk of SAEs in older adults was heterogeneous and strongly influenced by underlying conditions rather than age alone. Given the substantial risk of severe wild-type CHIKV infection in this age group, the PRAC lifted the temporary contraindication and confirmed a favourable overall benefit–risk balance, while emphasising the need for individualised risk–benefit evaluation in settings with meaningful exposure risk.

3.4. Fatal outcomes

Across the PRAC Article 20 assessment set, three fatal cases were identified, all in older men (77–88 years) with substantial multimorbidity. One case—an 84-year-old with encephalitis and vaccine-strain CHIKV detected in plasma and CSF—was assessed as probably related to IXCHIQ (WHO-AEFI “consistent”; WHO-UMC “probable”) and has been fully described elsewhere [21]. The remaining two deaths were assessed by the ARS as unlikely to be related to vaccination, a conclusion supported by the company's medical review.

The first involved an 88-year-old, profoundly frail man who had been

¹ It is important to emphasize that the figures on doses distributed and administered are based on estimates derived from commercial data sources. These estimates are subject to inherent limitations, including reporting delays, inconsistencies between distributed and actually administered doses, and variability in data collection practices across different healthcare systems. Additionally, the available data on age categories are limited to two groups: individuals below 65 years of age and those aged 65 years and above. More granular age stratification is currently not available from the collected data.

hospitalised five days before vaccination due to acute deterioration of long-standing neurologic and functional impairment, including severe mixed dementia, prior stroke with hemiparesis, dysphagia, and progressive weight loss. His admission status was consistent with a pre-final palliative phase. He subsequently developed aspiration pneumonia, deemed most plausibly attributable to underlying dysphagia and advanced frailty rather than vaccination.

The second case occurred in a 77-year-old man with advanced Parkinson's disease, prior myocardial infarction, and multiple chronic conditions. He experienced progressive functional decline beginning two days post-vaccination, likely exacerbated by dehydration in the context of limited home care. He was hospitalised on day 7 with aspiration pneumonitis, sepsis, renal failure, and decompensated Parkinson's disease, and died on day 14 despite supportive care. The clinical picture was most consistent with aspiration pneumonitis due to Parkinson-related oropharyngeal dysfunction, compounded by dehydration, rather than a vaccine-related mechanism. For both deaths, the PRAC classified causality as WHO-AEFI "indeterminate" and WHO-UMC "possible," reflecting substantial confounding and more plausible alternative explanations.

3.5. Quality and manufacturing data

As part of the Article 20 referral procedure, the EMA also reviewed manufacturing and quality data. All commercial doses administered worldwide up to April 2025 were derived from two validated drug substance batches. All distributed drug substance and product batches were analysed for potential links with AEs. Genetic stability was confirmed, potency and purity met specifications, and no correlation was observed between batch numbers and SAEs. Regulators concluded that the emerging reports were unlikely to be explained by a manufacturing defect or issue with manufacturing quality.

3.6. Off-label use

Within the serious ICSRs reviewed by the PRAC (EudraVigilance data cut-off 25 May 2025), the MAH categorised six of 28 serious cases (21%)

as off-label use because the individuals had underlying conditions such as immunodeficiency or malignancy. Importantly, the PRAC confirmed immunocompromised status in only three of these six cases; for the remainder, missing details (e.g., concomitant medications, malignancy status) precluded confirmation, leaving ambiguity regarding off-label use. These reports remain relevant for future emergency outbreak deployments, where off-label considerations may arise again in highly vulnerable patients.

3.7. PRAC recommendation, CHMP opinion, and final EC decision

On 10 July 2025, the PRAC concluded the Article 20 referral and confirmed a favourable benefit–risk balance for IXCHIQ, lifting the temporary contraindication in individuals aged ≥65 years. The committee recommended targeted updates to the Product Information (Fig. 1). These were incorporated into the Summary of Product Characteristics (SmPC) [16], clarifying that vaccination should be undertaken only when there is a significant risk of acquiring chikungunya and after individual risk–benefit evaluation (SmPC 4.4); reinforcing the contraindication in immunodeficient or immunosuppressed individuals (SmPC 4.3); and updating adverse reactions such as neurotropic events, syncope, thrombocytopenia, malaise, and decreased appetite (SmPC 4.8). The PRAC further advised issuing a Direct Healthcare Professional Communication and enhancing routine pharmacovigilance with focused monitoring of neurotropic, renal, and haematologic events, to be further characterised in planned post-authorisation studies.

In alignment with these recommendations, the Risk Management Plan was updated to include the identified risk of increased SAEs in adults ≥65 years with chronic medical conditions and the missing safety information for individuals aged 12–64 years with immune-dysregulating clinical conditions. Neurotropic events, renal impairment, thrombocytopenia, thrombotic microangiopathy, and off-label or immunocompromised use will be specifically monitored in forthcoming Periodic Safety Update Reports (PSURs), alongside additional phase 4 studies (VLA1553–401, VLA1553–406). Despite early serious reports predominantly in older adults with comorbidities, the Article 20 procedure reaffirmed a favourable benefit–risk profile for individuals with

SmPC 4.1
• Update to reflect the use of IXCHIQ in individuals aged 12 years and older, without a restriction in the elderly
SmPC 4.3
Remove the temporary contraindication in individuals aged 65 years and above. Clarify the wording of the contraindication in individuals who are immunodeficient or immunosuppressed.
SmPC 4.4
Add explicit guidance that vaccination should be given only when there is a significant risk of acquiring chikungunya and after careful consideration of individual risks/benefits. Include information on the serious reactions reported in elderly and in individuals with multiple chronic or uncontrolled conditions
SmPC 4.8
Include information on the serious reactions reported in elderly and in individuals with multiple chronic or uncontrolled conditions Add/adjust adverse reactions: • frequency "Not known": encephalitis, encephalopathy, aseptic meningitis; • frequency "Uncommon": syncope, thrombocytopenia, malaise, decreased appetite; • frequency "Rare": confusional state

Fig. 1. Amendments to the Product Information (PI) as advised by the PRAC, including amendments to sections 4.1, 4.3, 4.4, and 4.8 of the SmPC.

meaningful exposure risk. The CHMP adopted the PRAC's conclusions on 24 July 2025, recommending a variation to the marketing authorisation with revised Annexes I–III and updated scientific conclusions (Annex IV). On 12 September 2025, the EC issued the final Implementing Decision under Article 20, updating the Union Register accordingly. The revised EU SmPC now confirms the therapeutic indication for individuals ≥ 12 years and incorporates strengthened warnings/precautions, clarified contraindication wording, and updated adverse reaction information.

Summarised, the EU Article 20 outcome comprised: (1) removal of the temporary age-based restriction; (2) strengthened SmPC warnings, precautions, contraindications, and adverse reaction tabulation—aligned with the PRAC's scientific assessment, the CHMP's opinion, and the Commission's final decision.

3.8. FDA correspondence & US timeline (Apr–Aug 2025)

Post-licensure safety monitoring of IXCHIQ in the US has relied on the national reporting system (VAERS), co-managed by the CDC and FDA. As of December 2024, VAERS published 28 reports, including six serious reports from elderly travellers originating in the US. These cases were neurological or cardiac events that occurred 3–5 days post-vaccination in men ≥ 65 years, most (26/28) with multiple comorbidities. Five were hospitalised and subsequently recovered, and one met the FDA “other important medical event” criterion without hospitalisation [11]. These findings prompted a staged public-health response in 2025. Following ACIP deliberations in April, CDC introduced age ≥ 65 years as a precaution for IXCHIQ use—i.e., generally defer vaccination unless benefits outweigh risks—while continuing to recommend vaccination for travellers at substantial risk of exposure [11].

In May, following safety reports from La Réunion, the FDA and CDC issued a series of information requests to the MAH and recommended a temporary pause in administering the vaccine to individuals aged 60 and older, as additional serious reports were investigated. After a formal benefit–risk reassessment, FDA lifted the pause on 6 August 2025 and required updated US Prescribing Information (USPI) emphasising use primarily for individuals at high risk of chikungunya exposure and highlighting a potentially increased risk of serious events in adults aged 65 and older with chronic conditions [11]. The USPI was revised to strengthen warnings and precautions while maintaining availability; the FDA's IXCHIQ page shows the 6 August 2025 approval letter in the record, and the MAH issued a detailed press release describing the update on 7 August 2025.

However, within two weeks, the FDA suspended the license for IXCHIQ, effective 22 August 2025, citing ongoing safety concerns and halting the US distribution and sales. The FDA posted a public suspension notice and suspension letter, and Valneva confirmed the action in a press release on 25 August 2025. FDA's suspension of IXCHIQ's license cited four additional cases: three reports from Europe and one from the UK, which were all received directly from the local authorities and already captured in the global safety database. According to the MAH assessment, these cases did not materially alter the safety profile and therefore do not significantly alter the initial benefit–risk assessment. In addition to safety considerations, the FDA also highlighted the need for additional efficacy data, noting that the initial approval had been based on immunogenicity endpoints. This represented a shift in evidentiary emphasis compared with the original licensure framework and with other major regulators, such as the EMA and Health Canada, which continue to consider immunogenicity an acceptable surrogate for protection in this context. An editorial in *The Lancet Infectious Diseases* reaches a comparable conclusion, calling for additional publicly available details to enable independent appraisal of the evidentiary basis for the suspension [22]. As of 19 January 2026, following the FDA's August 2025 license suspension and subsequent IND clinical hold, the MAH voluntarily withdrew the U.S. biologics license application (BLA) and investigational new drug (IND) application for IXCHIQ.

4. Discussion

This integrated review, based on the EU Article 20 referral procedure and the company's analysis of 35 serious ICSRs, shows that post-marketing SAEs occurred predominantly in elderly individuals with multiple comorbidities, early after vaccination. A single probably related fatal encephalitis case with vaccine-strain detection defines the current upper bound of observed severity; the other two fatalities were indeterminate/possible amid compelling alternative explanations (aspiration, frailty). The PRAC/CHMP concluded that the benefit–risk remains favourable when used in individuals with meaningful exposure risk and after careful consideration of the potential risks and benefits, lifting temporary age measures and strengthening warnings/precautions alongside targeted pharmacovigilance. In the US, an information request (IR)-driven dialogue resulted in updates to the US Prescribing Information (early August 2025) emphasising individualised risk–benefit in older/multimorbid adults, followed by a temporary suspension later in August pending further evaluation.

4.1. Live-attenuated vaccine profile and clinical context

The early post-marketing safety pattern must be considered in the context of live-attenuated vaccine biology and the characteristics of the populations vaccinated during emergency use. Clinical trials included mostly healthy adults with limited representation of older individuals, and serious vaccine-related events were rare. Notably, the point estimate of vaccine-related SAE frequency in trials (5.7 per 10,000) appears similar to early post-marketing reporting (6.3 per 10,000); however, the small number of related SAEs in the clinical trial limits characterization and precludes robust subgroup comparisons. Consequently, post-marketing data primarily inform the clinical phenotype of rare events under real-world use. While early uptake was concentrated in older, multimorbid adults—limiting definitive attribution between age and comorbidity—the observed age-stratified differences in serious adverse event rates are consistent with an influence of age.

IXCHIQ's attenuation of the LR2006-OPY1 strain was achieved by reverse genetics resulting in a large deletion of 61 amino acids within the C-terminal part of the non-structural protein 3 (nsP3) of the replicase complex, resulting in restricted replication while preserving immunogenicity. Transient post-vaccination viremia has been observed, and clearance is expected as adaptive immunity develops. In specific individuals with clinically significant immune impairment or multiple comorbidities, prolonged viral persistence and/or atypical tissue distribution is biologically plausible and could contribute to chikungunya-like adverse reactions (CLARs), although the precise mechanism remains uncertain. While reversion to increased fitness appears highly unlikely given engineered attenuation and no change of deletion was detectable after serial passages on Vero cells, compensatory changes cannot be fully excluded for any replicating agent.

Systemic chikungunya-like reactions accounted for a substantial proportion of serious ICSRs and occurred during the expected early reactogenicity window of live-attenuated vaccines; one-third of these cases required hospitalisation, largely in individuals with marked frailty. Severe neurological events were uncommon, with one fatal encephalitis case confirmed as plausibly vaccine-related. Overall, the pattern of events is most consistent with increased host vulnerability during early real-world use rather than a clearly distinct new product-related syndrome; however, the small number of cases and the limitations of passive surveillance preclude definitive mechanistic attribution. Regulatory quality reviews did not identify manufacturing or batch-related issues, and the available data did not suggest a consistent cluster indicative of a novel systemic defect. This reinforces the need for clinical vigilance and individualised risk–benefit assessment in older and multimorbid adults who were not well represented in clinical trials.

4.2. Interpretation of post-marketing reports

Spontaneous post-marketing reports are central to signal detection but are not designed to quantify incidence or establish causality. For serious events, interpretation must consider the characteristics of both the numerator and denominator. Reporting is influenced by clinical vigilance, contextual factors, and the likelihood that medically significant events are more consistently captured than mild or expected reactions. At the same time, dose-based denominators may not accurately represent the population at risk for specific outcomes. Although most reported SAEs occurred in men, the absence of sex-stratified vaccine deployment denominators precludes determining whether this reflects differential uptake/reporting or a true difference in susceptibility. During the La Réunion emergency campaign, early vaccination efforts focused on older adults and individuals with chronic conditions. These groups have a higher baseline risk for serious medical events, increasing the probability of temporal—but unrelated—coincidences following vaccination. This context highlights the value of spontaneous reporting for generating hypotheses while underscoring the need for structured epidemiologic studies to assess causality and estimate incidence with appropriate control for underlying risk.

In evaluating neurological and systemic reports, limited diagnostic investigations constrained causal assessment, particularly in neurological cases without cerebrospinal fluid (CSF) analysis or virological testing. Most reports were classified by the PRAC as possible or indeterminate, reflecting the presence of comorbidities, incomplete clinical information, or plausible alternative diagnoses. Although temporal clustering after a live-attenuated vaccine is expected and can support—but not establish—causality, only 7 of the 28 serious cases assessed in the Article 20 review met criteria for probable or consistent vaccine-related events. The remainder remained indeterminate largely due to confounding factors and insufficient diagnostic data. These limitations highlight the need for high-quality case documentation, including standardised diagnostic work-ups, and underscore the value of integrating multiple data sources (e.g., EudraVigilance, VAERS, MHRA, Health Canada) to improve signal interpretation and contextualise emerging safety information. Taken together, these considerations underscore the need to interpret post-marketing safety data within their broader clinical and epidemiological context, rather than relying on temporal proximity alone.

4.3. Benefit-risk in the epidemiological context

When interpreted within the context of active chikungunya transmission, the benefit–risk balance of IXCHIQ remains favourable, particularly for older adults (≥ 65 years) who bear the highest burden of severe disease outcomes. During the 2024–2025 La Réunion outbreak, more than 47,500 reported cases and approximately 208 excess deaths were estimated in this age group [23,24], far exceeding the number of serious vaccine-associated events identified to date, including one fatal case with virological confirmation of vaccine-strain infection. These data highlight that, in endemic settings, the expected health impact of chikungunya substantially outweighs the observed post-marketing vaccine risks.

At the same time, the benefit–risk profile in non-endemic countries depends on the likelihood of exposure during travel and therefore differs from transmission settings. This underscores the importance of robust, context-specific modelling studies—separate from this work—to quantify the relative risk and benefit across age groups and epidemiological scenarios, including precise estimates for individuals aged ≥ 65 years. Strengthened warnings, clarified precautions, and targeted pharmacovigilance are therefore appropriate measures to optimise vaccine safety while preserving access for populations with meaningful exposure risk.

4.4. Who should consider IXCHIQ

Vaccination with IXCHIQ may be appropriate when the anticipated risk and clinical consequences of wild-type chikungunya virus infection outweigh the potential for serious vaccine-associated events, which to date have occurred predominantly in older adults with multimorbidity. IXCHIQ is contraindicated in individuals with immunodeficiency or immunosuppression, whether due to underlying conditions or medical therapy. These observations support continued emphasis in product information on careful recipient selection, particularly with respect to immune status and comorbidity burden. For individuals in whom IXCHIQ is contraindicated, alternative non-replicating chikungunya vaccine options, where authorised and available, may be considered. In addition to formal contraindications, substantially frail individuals or those with uncontrolled comorbidities may have an increased risk of adverse outcomes following any live-attenuated vaccination. Importantly, all vaccination decisions must adhere to national recommendations; within those frameworks, clinicians should perform an individualised risk–benefit assessment that considers exposure likelihood, clinical vulnerability, and immune competence rather than chronological age in isolation.

Based on current evidence and regulatory guidance, the following populations and scenarios represent circumstances in which IXCHIQ vaccination may reasonably be considered:

- Immunocompetent residents aged 12–64 years living in tropical or subtropical regions where chikungunya is endemic or where recurrent arboviral outbreaks occur.
- Travellers to outbreak or elevated-risk areas. Adolescents and adults aged 12–64 years travelling to regions with active chikungunya transmission and expected vector exposure (e.g., outdoor, rural, peri-urban, or unscreened/non-air-conditioned settings).
- Individuals with repeated or multi-destination travel. Adolescents and adults aged 12–64 years (18–64 in the UK and Brazil) whose travel patterns involve multiple destinations or recurrent trips to areas with elevated or outbreak-level transmission, resulting in cumulatively higher exposure risk.
- Long-term residence or deployment to areas with persistent transmission. Adolescents and adults aged 12–64 years relocating, working, or deployed to settings where chikungunya virus transmission is ongoing or recurrent.
- Field-based or operational personnel (military, humanitarian, diplomatic, NGO, public-health, or vector-control teams) operating in endemic or outbreak areas where exposure risk and limited access to timely care may be anticipated.
- Laboratory personnel with occupational exposure risk to chikungunya virus or infectious materials.
- Adults participating in local outbreak response campaigns when transmission is intense and timely access to acute medical care may be constrained.
- In regions with recurrent outbreaks, targeted vaccination of (healthy) higher-exposure groups may reduce the number of viremic infections and thereby decrease mosquito-mediated transmission, potentially providing indirect protection to those at highest risk, although the coverage required to materially affect transmission is not yet defined.

4.5. Clinical implementation and ongoing monitoring

Healthcare providers should conduct individualised risk–benefit evaluations that: (i) assess exposure likelihood based on the epidemiological situation at the destination, itinerary, and anticipated vector contact; (ii) consider host factors—including age ≥ 65 years and multimorbidity, which increase the risk of severe wild-type chikungunya and may also be associated with a higher vulnerability to serious vaccine-associated events; and (iii) review formal contraindications,

particularly immunodeficiency or immunosuppression.

If an individualised risk–benefit assessment for an older adult or a multimorbid adult supports vaccination, clinicians may consider optimising chronic conditions beforehand, staggering the administration of other vaccines when feasible, and providing clear guidance for the early post-vaccination period (days 2–7). This includes advising recipients to seek prompt medical evaluation if severe or unexpected symptoms occur.

Principles of risk-stratified vaccination—where clinical vulnerability and exposure risk jointly determine suitability—have long been applied to other live-attenuated travel vaccines (e.g., yellow fever vaccination in older adults or those with specific comorbidities). However, unlike yellow fever vaccine-associated viscerotropic disease (YEL-AVD), which shows a recognised age-associated pattern, no recurrent or age-specific syndrome has been demonstrated with IXCHIQ to date. Risk–benefit decisions must therefore rely on the current evidence base and jurisdiction-specific recommendations.

4.6. Strengths and limitations

A key strength of this review is the triangulation of evidence across multiple pharmacovigilance systems (EudraVigilance, VAERS, Health Canada, MHRA, MAH reports), regulatory assessments, and case narratives, with de-identified summaries provided in the Supplement. However, the findings are constrained by the inherent limitations of passive surveillance: variable reporting, incomplete diagnostics, limited follow-up, absence of robust denominators for specific subgroups, and confounding by indication. Estimates of doses distributed and administered are approximate and derived from commercial data, with age information available only for two strata (<65 and ≥ 65 years), limiting finer age-specific risk estimates. Finally, regulatory status evolves; clinicians should follow current agency communications and labels in their jurisdiction.

5. Conclusions

Chikungunya remains epidemic-prone, causing substantial morbidity and mortality in at-risk populations. Evidence from LAV biology, trials, post-marketing surveillance, and regulatory assessments supports continued availability of IXCHIQ with refined labelling and vigilant monitoring. Rare but serious adverse events were observed across age groups, with the most severe outcomes occurring in older adults with multiple comorbidities. Overall, the benefit–risk profile remains favourable for individuals at meaningful risk of exposure, provided that vaccination decisions are guided by an individualised assessment and accompanied by ongoing pharmacovigilance.

CRedit authorship contribution statement

Gerard Timmy Vondeling: Writing – original draft, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Julio Croda:** Writing – review & editing. **Tomas Jelinek:** Writing – review & editing. **George Kassianos:** Writing – review & editing. **Herwig Kollaritsch:** Writing – review & editing. **Liem Binh Luong Nguyen:** Writing – review & editing. **Guilherme Sousa Ribeiro:** Writing – review & editing. **Jonas Schmidt-Chanasit:** Writing – review & editing. **Stephen J. Thomas:** Writing – review & editing. **Zsuzsanna Unger:** Writing – review & editing. **Adrianne de Roo:** Writing – review & editing. **Eric Plennevaux:** Writing – review & editing. **Juan Carlos Jaramillo:** Writing – review & editing.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2026.128491>.

Data availability

All data relevant to this study are included in the article and its supplementary material.

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